

India Radio maintains many national and local AM stations. Apart from this advantage, AM radio can be detected and listened to using extremely simple equipment. With a strong enough signal, even a power source is unnecessary.



An early model transistor radio

Building an un-powered crystal set radio is a major childhood hobby activity for many hobby groups, and detailed instructions for this are commonly available. However, AM radio is held back by the fact that its signals are extremely sensitive to atmospheric conditions. Storms, solar flares, even wind, often disrupts good quality reception, confining AM to serve as a very utility oriented mode for radio broadcasting.

Frequency Modulation (abbreviated FM), is the undeniable star of today's radio broadcasting community. Crystal clear audio can be transmitted over acceptably long ranges via FM. FM signals are largely unaffected by atmospheric conditions. Requiring relatively lower bandwidth per channel, FM is now mostly transmitted in stereo, making it comparable to personal music players in sound quality.

Invented by Edwin H Armstrong in the 1930s as an answer to the problems inherent with AM, FM remained unpopular till the 1950s. Confined to frequencies between 42 and 50 MHz, FM remained largely a radio enthusiast's tool and the broadcasts were short, sporadic and experimental. The fact that the reception of FM signals required a different, special receiver also contributed to its lack of popularity. With the end of World War 2, more importance was given to improving the quality of radio transmissions. AM radio, by then suffering from a large amount of interference in the atmosphere, was begging for a